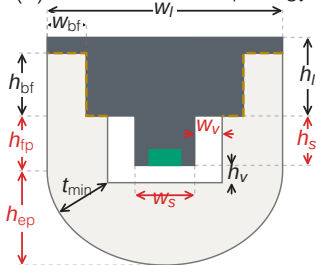


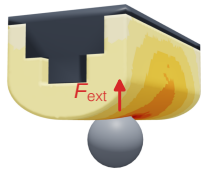
Rigid carrier
 Deformable pad
 LED / light source
 Bonding surface
 Spherical indenter
 Optical ray path

(a) Parameterized Morphology



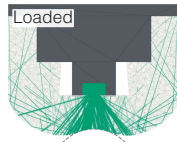
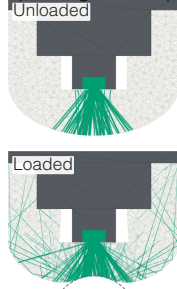
Design vector
 $\theta = [h_{fp}, h_{ep}, w_s, h_s, w_v]$

(b) Newton Soft Mechanics



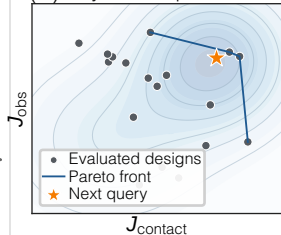
Mechanical Response $\rightarrow J_{contact}$

(c) OptiX Light Transport



Optical Response $\rightarrow J_{obs}$

(d) Bayesian Optimization



Propose next morphology
 θ_{i+1}

Update morphology and iterate